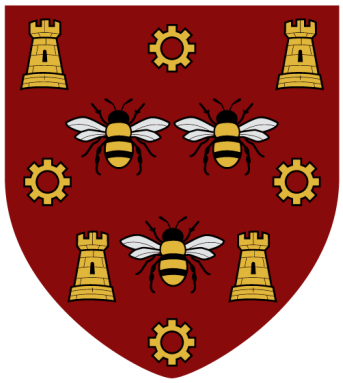
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Games Without Borders: Digital Play as a Global Communicative Infrastructure for Climate Policy Engagement

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ABSTRACT

Background: Digital games are increasingly used for civic purposes, yet questions remain about whether commercially mediated, non-deliberative interactions can constitute meaningful civic engagement. Existing research largely overlooks how entertainment platform infrastructures shape the democratic legitimacy and interpretability of user input at scale.

Objectives: This paper investigates whether ambient interactions within commercial games can meaningfully be understood as civic engagement, and under what conditions.

Methods: We analyse Play2Act, a climate-focused questionnaire embedded across 24 popular commercial mobile games as part of the GREAT project, which engaged over 181,000 participants from nearly every country. Drawing on civic media, platform studies and digital participation literatures, we examine the methodological, ethical and interpretive dimensions of embedding policy-relevant surveys in entertainment environments. The analysis focuses on how civic voice is structured by game design, platform affordances and datafication logics.

Results and Conclusions: Play2Act exemplifies what we term para-civic infrastructure: socio-technical systems that simulate civic input without supporting deliberation or accountability. Without evidence of participant intentionality or comprehension of the civic framing, these data are best understood as ambient civic signals rather than direct expressions of democratic will. We conclude that digital games are valuable diagnostic sites for understanding civic engagement at scale, and call for more dialogic and transparent design in future civic-technological interventions.

1 | Introduction

Can a fleeting moment in a mobile game become an act of civic expression? As digital infrastructures increasingly shape how publics engage with urgent issues such as climate change, this question takes on new urgency. Traditional models of civic participation, including petitions, consultations and deliberative forums, struggle to scale or reach across geopolitical divides. Yet commercial games, played by billions in diverse contexts, offer a global, under-examined medium for contact between institutions and publics.

Play2Act emerged within this tension: developed within the GREAT project, in partnership with UNDP, it tested whether policy-relevant surveys could be delivered through commercial mobile games without gamification or incentives. The campaign achieved large-scale global reach, including regions with limited civic infrastructure or media freedom, in

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ways that were unanticipated, raising questions about how games might be functioning as ambient infrastructures for civic voice. This paper treats Play2Act not simply as a data collection tool, but as a lens onto the para-civic possibilities of platform-based play. We argue that games can serve as low-friction, globally distributed systems for civic input, but that such potential is constrained by the structural logics of platforms, the ephemerality of interaction and the absence of reciprocal civic architectures.

This paper makes three contributions. It introduces para-civic infrastructure as a theoretically grounded framework for analysing game-based civic engagement that falls short of deliberative ideals. It demonstrates this through the most globally extensive empirical case study to date of survey-based civic engagement embedded in mainstream commercial games. And it provides a critical methodological account of the trade-offs involved in repurposing entertainment platforms for civic data collection, with recommendations for more accountable future designs.

2 | Theoretical Background and Key Concepts

2.1 | Civic Engagement, Civic Participation and Ambient Signals

Civic engagement refers broadly to individual and collective actions oriented towards public concerns or governance (Plantin et al. 2018). It spans a spectrum from voting and petitioning to informally sharing views. *Civic participation* is the active, narrower subset: acts that feed into recognisable political or policy processes with institutional recognition. The distinction matters here. Players responding to a climate survey inside Subway Surfers may have engaged civically in the broad sense, having encountered a public issue, but they did not participate civically in the narrow sense: their responses were neither solicited by a governing institution nor guaranteed to reach one.

Ambient participation names a mode of engagement that is backgrounded, incidental or embedded in non-civic contexts (Richardson 2020; Taylor-Smith and Smith 2016). Rather than requiring deliberate civic intent, it emerges within the flow of everyday digital activity, scrolling, tapping, playing, without the participant registering their action as political. Its observable outputs are *ambient civic signals*: patterned, measurable responses that may carry civic meaning but whose intentionality cannot be verified. In Play2Act, the survey completions are precisely this: real and geographically distributed, but epistemically indeterminate. Whether respondents understood their taps as climate advocacy or simply as a game prompt is something the dataset cannot resolve.

2.2 | Para-Civic Infrastructure and the Para-Civic Register

Para-civic participation describes modes of engagement that resemble civic action in form but lack its structural conditions: deliberation, accountability, institutional recognition and reciprocity (Butt et al. 2016). The prefix 'para-' signals a system that operates alongside civic infrastructure without constituting it. Deliberately designed civic games such as iCivics or Democracy 3 simulate civic reasoning (Kahne et al. 2009); para-civic participation, by contrast, arises in contexts not designed for civic purposes at all. Its defining characteristics are visibility without accountability; scale without deliberation and affective rather than procedural engagement. The *para-civic register* is the communicative mode associated with this: players are addressed as citizens-in-passing, invited to express views within a leisure frame, without the scaffolding of formal civic processes. A player of Avakin Life, a social life-simulation game, encountering a climate prompt during a loading screen is not navigating a deliberative forum; they are being hailed as a potential opinion-holder in the gap between gameplay moments.

2.3 | Low-Friction Design, Civic Friction and Extractive Participation

Low-friction design minimises the effort, time or expertise required to act. Low-friction civic entry points such as brief surveys, one-tap responses and embedded prompts lower participation barriers and reach audiences that demanding formats cannot (Korn and Voida 2015). Play2Act exemplifies this: completable in under 5 minutes, delivered within an already-open game, it accessed respondents across 228 countries and territories without asking them to leave their entertainment environment. *Civic friction*, by contrast, names deliberate design choices that slow engagement to invite reflection (Korn and Voida 2015). Play2Act's success depended on its low friction; the same quality limits its civic interpretability.

A related risk is *extractive participation*: processes in which user input is harvested without meaningful feedback, transparency about data use or pathways for co-governance (Andrejevic 2020; Salem 2016; Ming-Tak Chew and Wang 2021). *Reciprocal civic architectures*, by contrast, include feedback loops, accountability mechanisms and routes for participants to see how their input is used. Platforms such as Decidim or CrowdLaw attempt to build such reciprocity into civic digital tools (Thiel et al. 2019; Feddersen and Santana 2022). Play2Act lacked these features. Participants could opt in via email for a one-way campaign update, but could not shape survey questions, interrogate data usage or verify how findings were used by UNDP or partner studios. This absence is what distinguishes para-civic from genuinely civic participation.

2.4 | Participatory Infrastructure and Architectures of Participation

Participatory infrastructure refers to the technical and institutional systems through which participation is made possible, structured and constrained (Plantin et al. 2018; Constantinides et al. 2018). Infrastructure does not merely host participation; it determines what can be said, who is visible and how input is aggregated or ignored. *Architectures of participation* is the more

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granular concept, describing how specific platform designs configure what forms of engagement are possible, rewarded or visible (Mihailidis 2018; Chia and Ruffino 2022).

In game contexts this means asking whether players can express complex views or only binary choices, and whether civic prompts are woven into narrative or bolted on as external surveys. Minecraft's urban planning integrations use spatial building mechanics to allow nuanced community expression (Murphy 2024); Play2Act offered Likert-scale and multiple-choice inputs only, structuring engagement as preference-expression rather than deliberation. The architecture of each host game also mattered: Pokémon GO's location-based mechanic delivered survey prompts during real-world movement, while Subway Surfers' rapid arcade loop created cognitive conditions less suited to reflective responses. The choice of platform is therefore itself a political design decision.

Games are also *socially sticky* in ways that many civic platforms are not (Plantin and Punathambekar 2019): played daily, discussed among peer groups and integrated into routines and identity. This stickiness makes them effective vehicles for low-friction civic prompts, but means those prompts must compete with entertainment value, social interaction and achievement mechanics.

2.5 | Soft Participation, Platform Publics and Infrastructural Citizenship

Soft or indirect participation describes engagement that avoids explicit political framing, embedding civic expression within cultural or entertainment forms (Salem 2016; Hahn et al. 2016). In politically constrained contexts, soft participation may be the only available route. In Play2Act, the game environment offered communicative cover: the climate survey appeared within a leisure frame that reduced its perceived political stakes. *Play-as-response* names the specific mode that emerges when players respond to civic prompts within the flow of play, not as deliberate political actors, but as players whose momentary choices carry potential civic signal. This is qualitatively different from responding to an institutional survey: more spontaneous, more affectively inflected and less cognitively filtered.

Platform publics are the loosely assembled, globally dispersed collectives that form around shared digital participation without stable institutional membership (Yuan et al. 2019; Papacharissi 2022). This concept connects to Larkin's (2013) account of *infrastructural publics*, in which participation is not simply enabled or blocked by infrastructure but actively constituted by it; the systems through which people engage shape what kind of engagement is possible and what kind of public is produced. The related notion of *latent citizenship* (Isin 2008) further suggests that civic agency can exist in potential, present in a population but unrealised without the institutional conditions to activate it. Together these frameworks capture a mode of civic presence that is real but structurally constrained, which is precisely the condition platform-based participation tends to produce.

3 | Games, Civic Engagement and the Infrastructure of Digital Platforms

3.1 | Digital Games as Civic Instruments: Beyond the Serious Games Paradigm

The idea that digital games can serve as instruments for civic engagement is well-established within critical game studies. Games are recognised as complex cultural forms capable of modelling systems, provoking reflection and shaping attitudes (García-Peñalvo et al. 2021), and a significant body of research on serious and persuasive games has demonstrated their capacity to foster empathy, raise awareness and simulate policy scenarios through procedural rhetoric (Jacobs et al. 2020).

This tradition encompasses commercial off-the-shelf (COTS) games as well as purpose-built serious titles. Studies of Democracy 3, This War of Mine and Papers, Please have shown how commercial games can engage players with political and ethical dilemmas without being designed as civic tools (Shliakhovchuk 2024; Ryan et al. 2020), and research in Kenya and Nigeria demonstrates how COTS games can be mobilised for democratic learning and civic participation (Fisher 2020). This body of work is directly relevant to Play2Act's methodology, which deliberately avoided purpose-built serious games in favour of integrating prompts into existing commercial titles, a design choice grounded in evidence that COTS titles offer superior reach and cultural embeddedness (Yau et al. 2024). However, much of this work focuses on bespoke games with limited reach. The civic potential of mainstream mobile and casual games, those not explicitly designed around social or political themes, has been comparatively underexplored.

Commercial games can at times be more effective than serious games in communicating complex sociopolitical arguments, particularly when procedural and narrative elements align. The immigration simulation Undocumented uses procedural mechanics to generate identification with migrant experience in ways a purpose-built educational module is unlikely to achieve (Cleger 2015). Subway Surfers' parkour mechanic creates a sense of constant forward momentum that, when interrupted by a climate survey prompt, may produce a more spontaneous, more affectively charged response than a static webpage survey. Community PlanIt shows how civic games can structure participation through game mechanics, allowing community members to explore spatial trade-offs interactively (Gordon and Baldwin-Philippi 2014). These works collectively emphasise the civic affordances of games across contexts, from education and policy to grassroots mobilisation (García-Peñalvo et al. 2021; Szot 2024).

3.2 | Digital Platforms as Infrastructures of Participation

Recent scholarship has moved beyond viewing digital platforms as neutral delivery mechanisms, recognising them instead as infrastructures that actively mediate, shape and structure social, political and communicative life (Plantin and

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Punathambekar 2019; Hesmondhalgh et al. 2023). Rather than merely facilitating civic communication, platforms increasingly constitute the conditions under which civic discourse becomes possible or constrained (Plantin et al. 2018; Constantinides et al. 2018). Within this frame, games are architectures of participation: spaces in which civic engagement is structured through mechanics, interfaces and platform logics. When Niantic integrated Play2Act into Pokémon GO, participation was shaped not only by players' climate attitudes but by Niantic's proprietary advertising infrastructure, the timing of delivery and the visual aesthetics of the interface. The civic signal was inseparable from these commercial and technical mediations. Scholars have highlighted how dominant platform architectures may undermine civic voice, reconfiguring users as consumers rather than citizens, and prompting calls for counter-platforms that resist corporate logics (Butt et al. 2016). In the context of games-based civic engagement, this demands critical reflection not only on game content but on the infrastructural conditions, including distribution platforms, data governance and interaction design, through which civic participation is enabled or foreclosed.

3.3 | Cross-Border Communication and Geopolitical Asymmetries

The idea of the internet as a borderless space has long dominated narratives of digital globalisation. Yet cross-border communication is mediated by technological infrastructures, cultural dynamics and regulatory regimes that reproduce divisions rather than dissolve them (Hafez et al. 2022; Mattelart et al. 2024). Global digital platforms remain subject to territorial borders, censorship regimes and infrastructural disparities that shape what information flows, where and for whom (Tawil-Souri 2015; Lee 2020). Authoritarian states deploy sophisticated digital controls to regulate online discourse (Sinpeng 2020), yet certain digital formats, particularly games, may act as semi-anonymous channels capable of circumventing state filters through oblique content or interactive ambiguity (Hahn et al. 2016). These platforms enable 'soft' or indirect participation, blending entertainment with subtle forms of resistance or engagement (Salem 2016; Ming-Tak Chew and Wang 2021). Understanding games as communicative infrastructures in these settings requires attention to both their potential and their limits: how they enable indirect civic meaning-making, but also how platform logics, content moderation and geo-blocking may re-inscribe existing asymmetries.

3.4 | Climate Engagement and the Role of Games

Traditional climate engagement strategies have leaned on structured formats such as citizen assemblies, deliberative forums and policy consultation surveys. While these hold normative value in democratic theory (Blue 2015; Boswell et al. 2023), they require time, facilitation and a degree of civic confidence and may inadvertently exclude those less institutionally connected or who view climate discourse as the domain of experts (Polletta 2018; Romsdahl 2020). The spectrum of climate-related games is growing, with over a 100 identified across genres and formats, although many remain focused on education rather than civic or policy outcomes (Gerber et al. 2021). The empirical evidence base on their effectiveness in large-scale, cross-cultural or policy-relevant contexts remains thin (Fernández Galeote and Hamari 2021). Play2Act enters this landscape by embedding civic prompts into existing digital game spaces, reframing participation as something that can be ambient, incidental or even playful and raising important questions about how climate narratives are mediated through popular culture infrastructures.

Mobile and casual games function as everyday media environments, ubiquitous, habitual and often ambient. Titles such as Subway Surfers (downloaded over 3 billion times) or Pokémon GO (over 80 million active monthly players at peak) reach populations that deliberative civic platforms cannot. They circulate via app stores, peer recommendations and platform economies, often sidestepping regulatory controls not by subversion but by appearing apolitical or playful, which is precisely what made Play2Act's global reach possible, and precisely what complicates reading it as civic.

4 | Materials and Methods

4.1 | What Is Play2Act?

Play2Act is a climate-focused survey initiative developed within the GREAT project in partnership with UNDP. The survey was deployed across 24 popular commercial mobile games, including Pokémon GO! (Niantic), Subway Surfers (SYBO), Beatstar (Space Ape Games), Avakin Life (Lockwood), Angry Birds 2 and Dream Blast (Rovio), RuneScape (Jagex), Love & Pies (Trailmix), Nitro Nation (Creative Mobile) and Fishing Clash (Ten Square Games). These games were selected for their membership of, or alignment with, the Playing for the Planet Alliance, a UN Environment Programme initiative bringing together major gaming studios to integrate environmental themes and reduce the sector's carbon footprint. The 24 participating studios collectively reached approximately 80 million players per week, representing a broad cross-section of genres: location-based augmented reality (Pokémon GO), casual endless-runner (Subway Surfers), rhythm/music (Beatstar), social life simulation (Avakin Life), competitive strategy (RuneScape, Fishing Clash) and racing (Nitro Nation). This genre diversity enabled comparison of how different gameplay contexts shaped survey engagement.

The survey comprised 11 items designed collaboratively with UNDP and academic partners, drawing on prior large-scale engagement initiatives including Mission 1.5 and the Peoples' Climate Vote (de Jong and Mneimneh 2021; Parry et al. 2021). All items used closed-response formats, multiple-choice or Likert scale, to maximise comparability across languages and minimise cognitive load. The questionnaire was designed to be completable in under 5 minutes, with no more than one or two items per screen. Items were structured around UNDP climate policy dilemmas, asking players to express attitudes and self-reported intentions rather than to demonstrate knowledge; representative questions are shown in Table 1. Demographic items captured gender, age and education leaving age. No personally identifiable information was collected; responses were anonymised prior to analysis.

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TABLE 1 | Representative survey questions in Play2Act.

Question	Response type	No. of options
1. Do you think video games can play a role in solving climate change or environmental issues?	Single choice	3
2. How can video games best contribute to fighting climate change and preserving nature?	Multi-select	5
3. Have you ever played games that include green messages or content about climate change or environmental causes?	Multi-select	4
4. After playing green game content, how did you feel about climate change and nature issues?	Likert	7
5. After playing green game content, have you changed anything you do in your day-to-day life?	Multi-select	6
6. From the following options, what would motivate you or has motivated you the most to play games with green messages or content?	Multi-select	6
7. How do you feel about climate change and nature issues?	Likert	7
8. Have you heard of any of the following agreements or plans?	Multi-select	5
9. What is your gender?	Banded range	4
10. How old are you?	Categorical	4
11. How old were you when you left education?	Banded range	4

Questions avoided technical policy vocabulary and were translated into 10 languages (Arabic, English, French, German, Italian, Japanese, Mandarin, Portuguese, Russian and Spanish) by local partners. This framing also minimised perceived political risk, a design choice that may have enabled high completion rates in politically restricted environments but equally means the items cannot distinguish genuine climate concern from socially desirable responding.

Open-ended questions were excluded based on evidence from the GCS1 pilot study (Harris et al. 2026), which found that free-text entry in game environments consistently produced low completion and high item non-response, reflecting the cognitive mismatch between text composition and mobile play interaction patterns, with comparable findings reported in Mission 1.5 and the Peoples' Climate Vote (de Jong and Mneimneh 2021; Parry et al. 2021). The absence of open-ended items means the dataset captures what players choose among options provided, but not why.

4.2 | How Was Play2Act Embedded Within Games?

To avoid disrupting gameplay or narrative, survey invitations were delivered through external communication tools, including in-app banners, loading screen prompts, message overlays and in some titles proprietary advertising infrastructure. Participation was therefore adjacent to gameplay rather than integrated within it (Figure 1): unlike civic game designs where player choices generate civic data within gameplay, such as SimCity land-use decisions as planning preferences, Play2Act operated as a contextually embedded but technically separate survey. The civic prompts did not affect game state, player scores or in-game rewards; participation was entirely voluntary, with no financial or in-game incentives and the survey could be dismissed without penalty. In practice, a player encountering Play2Act would see a prompt during a natural pause in gameplay, tap to enter the survey, respond to a short sequence of questions, and return immediately to the game. This approach prioritised minimal friction and organic uptake (Keusch et al. 2019). This design preserved the integrity of the civic signal at the cost of some degree of integration with gameplay flow. While participation carried no game-mechanic incentives, the game environment itself constitutes an implicit motivational frame; players were already in a context of voluntary, leisure-oriented digital activity and this contextual embeddedness is treated throughout as analytically significant rather than incidental to the civic signal produced.

4.3 | Sampling, Reach and Demographic Profile

Out of 934,224 sessions across 228 countries and territories, 204,796 users responded to at least one survey question and 181,087 completed the full questionnaire, a completion rate of approximately 19.4% of initiated sessions, reflecting the voluntary, non-incentivised model. The sample included participants from nearly all UN-recognised nations, including 70 Least Developed Countries or Small Island Developing States, with notable representation from Latin America, Southeast Asia and Sub-Saharan Africa. Participant location was inferred automatically via IP-based geolocation, which enabled broad geographic mapping without user input but provided no additional demographic detail. The demographic distribution skewed younger and more male; approximately 70% of respondents were under 36, and 51.3% identified as male, with 36% female and a notable proportion preferring not to disclose gender, consistent with global mobile gaming patterns (Jäckle et al. 2019), and voluntary participation introduced selection effects: respondents were more likely to be digitally literate, have stable mobile data access

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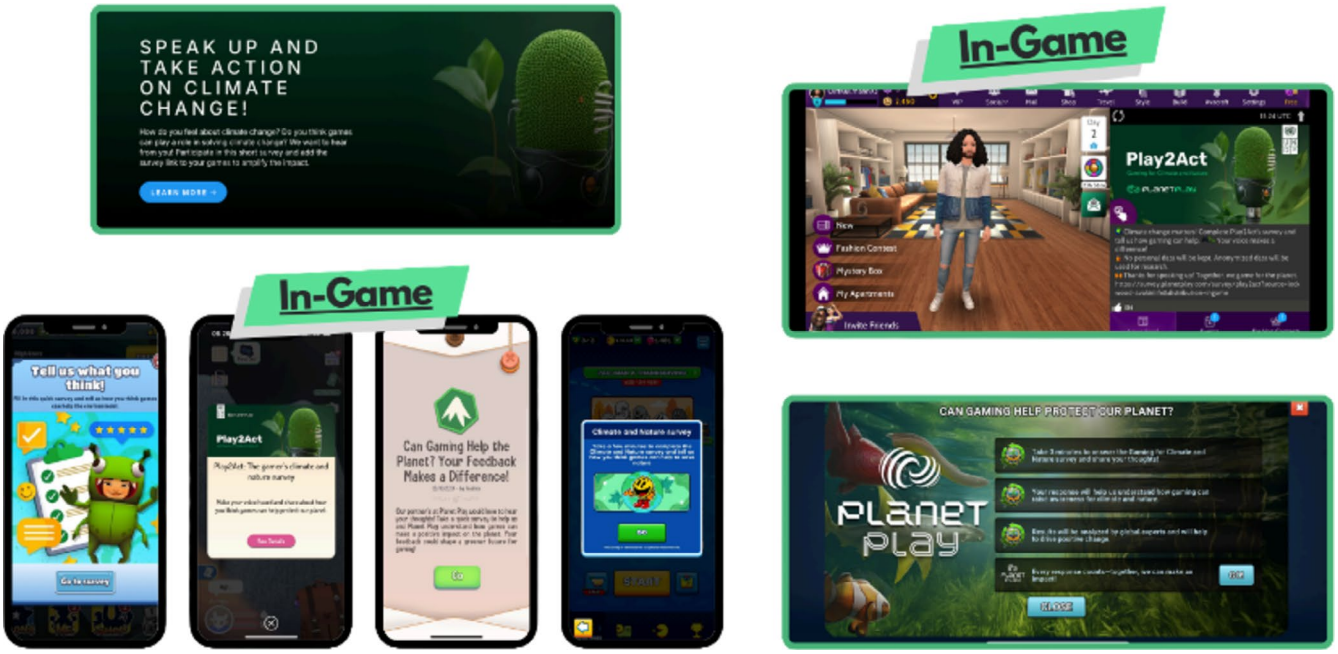


FIGURE 1 | Illustration of the placement of survey prompts within Play2Act.

and possess pre-existing interest in climate issues, all constraints on the generalisability of findings. Gender differences were consistent throughout the dataset: female respondents showed higher hope and motivation across climate attitude measures, while male respondents showed more informational confidence alongside elevated scepticism, a pattern that recurred across regions and game titles and warrants attention in future civic game design.

4.4 | Ethical Framework

Participants were recruited through in-game prompts clearly communicating the survey's purpose, anonymity and voluntary nature. Consent was secured through platform-native mechanisms compliant with GDPR and aligned with Horizon Europe and UKRI ethical frameworks; the lead academic institution obtained ethical approval with specific attention to data ownership, user privacy and responsible research practices in commercial environments.

4.5 | Analytical Approach

The dataset was analysed using descriptive statistics, cross-tabulations and regression models to explore correlations between climate attitudes and variables including region, game genre and user demographics. Platform-specific comparisons assessed how different gameplay mechanics influenced response patterns. Chi-square tests (with Bonferroni post hoc corrections where applicable) were used to assess the statistical significance of associations between regional, genre and demographic variables and response distributions. ANOVA was employed to examine relationships between Climate Risk Index scores and attitude patterns. All findings are correlational; no causal claims are advanced.

5 | Results

The findings are presented across three dimensions: patterns of geographic reach; variation across game genres and mechanics and the interpretive implications of participation in politically constrained contexts. Throughout, the analysis draws on the conceptual framework established above, interpreting the data through the lens of ambient participation, para-civic infrastructure and architectures of participation, to evaluate how structural conditions actively shape these patterns.

5.1 | Geographic Reach and Participation as Ambient Civic Signal

Across the campaign, Play2Act was launched in 934,224 game sessions across 228 countries and territories. Of these, 204,796 players responded to at least one question, and 181,087 completed the full survey, illustrating the capacity of mobile-first game delivery to circumvent traditional barriers to participation such as literacy, political interest or institutional trust.

Play2Act recorded high completion rates in countries with limited political freedoms, including China, Iran and Myanmar, with respondents completing multiple items consistently. While notable, these patterns must be interpreted with caution: in environments where formal political expression is restricted, alternative explanations include the appeal of anonymous digital platforms, variation in censorship enforcement and network access (Lin and Zhang 2020). The convenience sampling method and voluntary participation mean respondents may not reflect the broader public or politically disengaged groups. Rather than interpreting these data as straightforward evidence of democratic empowerment, they are best understood as ambient civic signals: patterned, observable, but epistemically indeterminate. The game environment likely provided a plausibly apolitical

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frame that reduced perceived risk. Responses from restricted contexts may reflect genuine climate concern, but may equally reflect the lower perceived stakes of responding within a leisure format, functioning as communicative cover that a labelled 'Government Survey' would have suppressed. High completion rates may in some cases reflect gamified motivation rather than civic resonance. Interpreting these engagements as civic participation requires stronger evidence of intentionality and comprehension that this dataset cannot provide.

Regions in Eastern Europe and parts of South Asia, despite high mobile penetration and active gaming communities, showed unexpectedly low completion rates. This asymmetry points to infrastructural and cultural contingencies mediating digital participation: differences in platform availability, cultural attitudes towards survey participation, prior exposure to green gaming content and variation in how the survey was surfaced within each game's interface. These patterns are consistent with the broader CRI analysis: countries facing the highest climate vulnerability (CRI < 10) showed the greatest scepticism and uncertainty about games' civic potential, 29% no, 35% not sure, compared with 52%–53% yes in lower-risk bands, suggesting that ambient civic infrastructure is not only unevenly distributed geographically, but that its reach is weakest precisely where climate stakes are highest (Table 2). These asymmetries are not noise: they provide evidence that para-civic infrastructure is shaped by platform capitalism's own geography of visibility and reach, systematically surfacing some voices while muting others.

5.2 | Genre, Mechanics and Patterns of Climate Engagement

Initial analysis suggests game genre and mechanics correlated with distinct patterns of climate-related engagement. Chi-square tests confirmed that these genre differences are statistically significant ($\chi^2 = 559.001–1671.279$ across attitudinal measures, all $p < 0.001$), though Spearman's correlations were consistently weak ($\rho = -0.023$ to 0.094), indicating that while genre is a statistically reliable predictor, its practical effect size is modest. The following patterns are therefore interpreted as tendencies rather than strong determinants; these tendencies and the specific game-level patterns are summarised in Table 3.

Pokémon GO, a location-based augmented reality game in which players physically traverse real-world environments to discover virtual creatures and resources, reported the highest levels of climate optimism (59% believing games can address climate change, compared to the dataset mean of approximately 51%), environmental behavioural intent (41%) and perceived knowledge gain (22%). The location-based mechanic may plausibly connect players to their physical environment in ways that make environmental concerns more proximate and affectively charged, consistent with research linking place-based interaction and augmented reality to enhanced environmental engagement (Whalen and Kijne 2019; Neset et al. 2020). When a Pokémon GO player pauses their walk through a local park to answer a climate survey prompt, they are literally in a natural environment, which may activate environmental affect in ways a player of a purely virtual game is not. This suggests that spatial embeddedness in game design may function as a form of civic friction, one that slows attention and opens space for reflection.

F1 Clash, a mobile racing management game simulating engineering decisions about car performance, showed elevated rates of climate motivation and support for mitigation policies (56% on Q1, with low scepticism), plausibly reflecting its implicit engagement with energy and technology themes as Formula 1 is

TABLE 2 | Regional participation patterns and climate risk profile.

Region	Believe games can help (Q1)	Hopeful after green content (Q4)	Climate risk index (CRI) profile	Notes
Africa	57%	22% hopeful; 25% informed; 20% sceptical	High vulnerability (Asia/Africa highest CRI scores)	Strongest Q1 belief (57%); LDCs show 61% belief, 12% higher than non-LDCs; SIDS 55% belief; elevated scepticism (20%) may reflect mistrust despite high climate exposure
Americas	50%	24% hopeful; 28% informed; 9% sceptical	Lower CRI (lower vulnerability)	Lowest scepticism of any region (9%); most emotionally positive post-engagement, although Q1 belief more moderate than Africa or Asia
Asia	56%	18% hopeful; 28% informed; 24% sceptical	Highest regional CRI scores	High scepticism despite strong engagement; most policy-aware region (Q8); cognitive-emotional disconnect
Europe	46%	21% hopeful; 26% informed; 13% sceptical	Lower vulnerability	Highest 'No' response on Q1 (26%); mirrors high-income scepticism pattern
Oceania	43%	22% hopeful; 24% informed; 11% sceptical	Lower vulnerability	Lowest Q1 confidence; highest uncertainty (31% not sure); weakest motivation (23% 'none of above')

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TABLE 3 | Game-level engagement patterns across key attitudinal measures.

Game/Genre	Believe games can help climate (Q1)	Notes
Pokémon GO (AR/location-based)	59%	Highest across all measures; lowest scepticism (4%). Location-based mechanic may prime environmental affect
F1 Clash (racing/strategy)	56%	Elevated climate motivation; implicit energy/technology themes plausibly explain engagement
Beatstar (rhythm/music)	51%	Moderate engagement; high uncertainty (30% not sure on Q1)
Subway Surfers (arcade/endless runner)	45%	Lowest Q1 belief; highest 'not sure' on green content (30%). Rapid arcade loop may reduce cognitive availability for civic prompts
Xbox Insider (mixed)	49%	Highest disengagement; 13% scepticism on Q4

Note: All associations are correlational.

actively involved in sustainability transitions. Players managing engine types, fuel loads and performance upgrades may be primed to engage with real-world analogues of these decisions. These trends are consistent with research linking strategic gameplay to reflection and decision-making (García-Peñalvo et al. 2021), but remain correlational.

Conversely, arcade-style games such as Subway Surfers showed lower completion rates and weaker pro-environmental dispositions (45% on Q1, and the highest 'not sure' rate on green content at 30%). Subway Surfers' gameplay loop prioritises rapid, automatic response and flow states that may be more disrupted by a text-based survey prompt than the slower, more reflective modes of engagement associated with strategy or AR games. Players in flow states may be less cognitively available for civic prompting, suggesting that the civic affordances of games are not merely a function of content or environmental themes, but of interaction tempo and cognitive demand.

These genre-specific findings connect directly to architectures of participation: different games embed different rhythms of attention, interaction and social context, shaping the conditions under which civic prompts can be received and processed meaningfully. A survey embedded in a loading screen creates a qualitatively different participatory opportunity than one delivered in a social multiplayer context. These distinctions deserve systematic investigation; current findings do not permit causal inference, and self-selection into genres is a significant confound. Without mechanisms for feedback, reflection or sustained involvement, it is also not possible to assess whether attitudes shifted or engagement extended beyond the moment of interaction.

5.3 | Green Game Engagement and Behaviour Change

A key finding, with direct implications for how these data should be interpreted, concerns the relationship between exposure to green game content and real-world behaviour change. Chi-square tests found no statistically significant association between green game engagement and any of the four behavioural change categories tested (Table 4). These null findings are not a failure of the campaign; they are an important epistemic constraint on the civic claims that can be made for ambient participation. The absence of behaviour change effects reinforces the argument that Play2Act data should be treated as ambient civic signals, indicators of distributed climate sentiment, rather than as evidence of civic impact or democratic efficacy.

5.4 | Policy Framework Awareness

A further dimension of the findings, with direct bearing on how the civic meaning of Play2Act data should be assessed, concerns participants' awareness of the specific international climate frameworks that the survey was designed, in part, to inform. Over 80% of respondents globally were unfamiliar with Nationally Determined Contributions (NDCs), the Kunming-Montreal Global Biodiversity Framework and National Biodiversity Strategies and Action Plans (NBSAPs), even among respondents who expressed strong environmental concern and who had previously engaged with green game content (Table 5). This pattern held across regions, income levels and CRI bands, and was consistent regardless of game title.

This finding is conceptually significant. Play2Act captured widespread climate sentiment without any corresponding policy literacy; respondents could express concern about climate change and support for environmental action while having no knowledge of the specific frameworks their responses were intended to inform. This disconnect between civic sentiment and policy awareness is precisely what the concept of ambient civic signals is designed to capture: patterned, observable, but epistemically indeterminate. It does not invalidate the dataset, but it does substantially constrain the claims that can be made about its democratic significance.

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6 | Discussion

6.1 | What Is Civic Participation in Platform-Based Games?

While Play2Act demonstrates the feasibility of embedding surveys into commercial game environments at scale, its civic implications must be interpreted with caution. Participation here is defined not simply as interaction or behavioural compliance, but as public-oriented action involving some recognition, by participant or system, of political or civic significance: expressive acts, input into decision-making or contribution to the public sphere (Plantin et al. 2018; Constantinides et al. 2018).

Play2Act operated at the level of ambient, low-friction interaction and lacked any direct inquiry into how participants perceived their own engagement. Did players understand the activity as contributing to public discourse or policy? Did they see themselves as engaging politically, or simply responding to a game prompt? The dataset cannot resolve these questions, and interpreting such interactions as expressions of civic concern risks overextension (Taylor-Smith and Smith 2016). The null findings on behaviour change, and the near-universal unfamiliarity with the policy frameworks the survey was designed to inform, reinforce this caution empirically: civic sentiment and civic knowledge are not the same thing, and neither is equivalent to civic participation. Ambient participation is not inherently without value: it may surface latent civic attitudes that deliberate consultation would not reach, particularly among populations who distrust formal political institutions. But treating ambient engagement as equivalent to deliberate civic voice collapses a distinction that matters for democratic legitimacy.

TABLE 4 | Chi-square tests for association between green game content exposure and self-reported behaviour change ($n = 125{,}821$).

Behaviour category	χ^2 statistic	p	Significant?
Changed purchasing to greener options	1.166	0.280	No
Reduced environmental impact (daily habits)	2.596	0.107	No
Participated in environmental campaigns	0.390	0.532	No
Informal social/community action	1.430	0.232	No
Reported no influence on behaviour	6.895	0.009	Yes (marginal)

6.2 | Para-Civic Infrastructure: Risks and Limits

Play2Act operated within a 'para-civic' register: a mode of engagement that mimics civic forms without offering their institutional depth. The campaign reached millions through infrastructures designed for entertainment and monetisation, not deliberation or accountability, exposing the paradox of attempting civic engagement through systems that are structurally opaque, commercially driven and not designed for reciprocal governance.

This foregrounds a broader asymmetry: who controls the narrative of civic input? Who benefits from the data? How can legitimacy be claimed where consent is limited to participation, not co-governance (Butt et al. 2016; Plantin and Punathambekar 2019; Hesmondhalgh et al. 2023)? In Play2Act, data were held by PlanetPlay and accessed by UNDP, institutions with legitimate mandates but not democratically accountable to the player-respondents. Aggregated findings were shared with game studios and used in UNDP climate policy dialogue, but individual participants had no mechanism to challenge, contextualise or opt out of this downstream use post hoc. Participants could opt in via email for a one-way campaign update, but could not shape survey questions, interrogate data usage or influence how findings would be interpreted by institutions. This absence of dialogic infrastructure is what distinguishes para-civic from genuinely civic participation.

6.3 | Power, Geopolitics and the Limits of Infrastructural Reach

The global asymmetries in Play2Act data, notably low uptake in Eastern Europe and parts of South Asia despite high gaming activity, are not merely operational curiosities. They reveal how commercial platform geographies re-inscribe geopolitical

TABLE 5 | Awareness of international climate policy frameworks among Play2Act respondents.

Framework/Agreement	% Aware (Global)	% Unaware/ Not heard of	Notes
Nationally determined contributions (NDCs)	< 20%	> 80%	Low awareness even among respondents expressing strong climate concern
Kunming–Montreal global biodiversity framework	< 20%	> 80%	Consistent across regions including high climate-risk countries
National biodiversity strategies and action plans (NBSAPs)	< 20%	> 80%	Lowest recognition of the three frameworks tested

Note: Awareness consistently low across all regions and demographic subgroups.

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inequalities within ostensibly global civic systems. The CRI analysis adds a further structural dimension: the populations facing the greatest climate risk are, in Play2Act, the most sceptical about games' civic potential and the least reached through its infrastructure. Countries with CRI scores below 10, the most climate-vulnerable, showed the highest rates of disbelief (29%) and uncertainty (35%) about gaming's role, while Least Developed Countries, despite showing the strongest general belief (61%) in games' potential, also displayed the lowest policy awareness and highest emotional uncertainty after engagement. This pattern suggests that ambient civic infrastructure amplifies voices that are already relatively empowered, those with stable digital access, cultural familiarity with survey formats and insulation from immediate climate consequences, while underrepresenting precisely those whose perspectives most urgently need surfacing. It is not just who participates but which platforms' algorithms surface the civic prompt, and in which languages, that determines whose voices are counted. The high response rates from authoritarian contexts must be read alongside the awareness that engagement may be shaped by platform mediation rather than genuine civic expression: governments may strategically embrace digital participation frameworks to expand surveillance or simulate openness without ceding control (Salem 2016).

6.4 | Towards More Accountable Civic Game Design

Researchers and designers must move towards more reflexive and accountable civic infrastructures. In-game dashboards showing aggregate results, participatory design stages to shape survey framing or post-campaign messages detailing how civic inputs translated into action would begin to reposition participants as publics with epistemic and ethical standing rather than data points. These mechanisms were absent from Play2Act's first phase but represent concrete recommendations for successor initiatives, including a second phase that offers an opportunity to test reciprocal features at scale.

One implication of these findings is that more participatory and dialogic design approaches, such as co-design, may help address the absence of feedback and accountability observed here. Within the GREAT project, Yau et al. (2024) tested such approaches with citizens and policy stakeholders, finding that early co-design can surface assumptions about what counts as a civic question, diversify the policy issues addressed and build participant trust. Future iterations of Play2Act should embed co-design before survey questions are finalised, rather than treating civic consultation as a downstream activity. Participatory depth can also be layered into ambient formats through optional narrative branches, short open-text prompts or follow-up invitations that allow users to reflect further.

The gender patterns in the data add a further design implication. The consistent finding that female respondents showed higher hope and motivation while male respondents showed more informational confidence and scepticism suggests that civic prompts embedded in games are experienced differently across gender lines, and that future designs should consider how framing, tone and interactivity might be calibrated to engage these different orientations rather than defaulting to a single civic register.

7 | Conclusion

Can a fleeting moment in a mobile game become an act of civic expression? The Play2Act case study suggests that it can, but with profound caveats. Three arguments have been advanced here. Commercial games constitute a form of para-civic infrastructure: globally distributed, culturally embedded systems that can surface ambient civic signals at unprecedented scale, but that lack the deliberative, reciprocal and accountable features of genuinely civic institutions. The civic meaning of ambient participation is epistemically indeterminate without evidence of participant intentionality, a constraint this study cannot overcome, and which all future game-based civic research should acknowledge explicitly. And the genre and mechanics of host games systematically shape the conditions of civic engagement, making the choice of platform itself a political design decision.

Participation was shaped by algorithmic visibility, survey design constraints and the asymmetries of platform capitalism. Engagement appeared in fragmented, fleeting and datafied forms, resonating with the notions of latent citizenship (Isin 2008) and infrastructural publics (Larkin 2013): both describe how civic meaning can coexist with structural conditions that prevent its full realisation. The null findings on behaviour change and the widespread unfamiliarity with the policy frameworks these data were meant to inform give this observation empirical grounding: Play2Act's respondents were civically present but not civically equipped, their participation structurally ambient rather than deliberatively embedded. Play2Act's respondents exemplify both, as potentially civic actors operating through an infrastructure not designed to honour their agency.

Three priorities for future research emerge: longitudinal studies to trace whether platform-based civic engagement influences attitudes or behaviours over time; genre-specific experiments investigating how different mechanics such as narrative immersion or competitive play shape civic affect and agency; and civic co-design trials bringing policy actors, developers and publics into direct collaboration on participatory systems. To these we add a methodological priority: future studies should include direct measures of participant intentionality and comprehension, through short exit questions, deliberation prompts or post-session reflections, to address the fundamental epistemic gap that limits this study's civic claims.

For practitioners, Play2Act serves as both proof-of-concept and a cautionary tale. It shows that civic engagement can emerge not only in town halls or voting booths, but in the quiet intervals of digital play. Yet recognising such activity as civic demands a deeper interrogation of its conditions of emergence, not just creativity, but accountability.

Play2Act offers less a blueprint than a warning shot. If games are becoming infrastructures of civic voice, we must ask: voice to whom, on whose terms and towards what end? The challenge ahead is to imagine civic infrastructures that are not only scalable, but dialogic, ethical and capable of rendering both publics

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and institutions visible to one another in meaningful, durable ways.

Author Contributions

Rebecca Harris: writing – original draft, writing – review and editing, project administration, investigation. **Anchal Garg:** data curation, validation, formal analysis. **Joost Schuur:** data curation, resources, investigation. **Paul Watson:** supervision. **Jude Ower:** data curation, resources, investigation. **Celestine Iwendi:** supervision. **Pradeep Hewage:** data curation, validation, formal analysis. **Paul Hollins:** methodology, project administration, funding acquisition, conceptualization, investigation.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in Zenodo at https://www.zenodo.org, reference number 101094766.

Peer Review

For transparency, the peer review documents associated with this article are available at https://doi.org/10.1002/jcal.70290.

References

Andrejevic, M. 2020. "Data Civics: A Response to the "Ethical Turn"." *Television and New Media* 21, no. 6: 562–567.

Blue, G. 2015. "Public Deliberation With Climate Change: Opening Up or Closing Down Policy Options?" *Review of European, Comparative & International Environmental Law* 24, no. 2: 152–159.

Boswell, J., R. Dean, and G. Smith. 2023. "Integrating Citizen Deliberation Into Climate Governance: Lessons on Robust Design From Six Climate Assemblies." *Public Administration* 101, no. 1: 182–200.

Butt, D., S. McQuire, and N. Papastergiadis. 2016. "Platforms and Public Participation." *Continuum* 30, no. 6: 734–743.

Chia, A., and P. Ruffino. 2022. "Special Issue Introduction: Politicizing Agency in Digital Play After Humanism." *Convergence* 28, no. 2: 309–319.

Cleger, O. 2015. "Procedural Rhetoric and Undocumented Migrants: Playing the Debate Over Immigration Reform." *Digital Culture & Education* 7, no. 2: 19–39.

Constantinides, P., O. Henfridsson, and G. G. Parker. 2018. "Introduction—Platforms and Infrastructures in the Digital Age." *Information Systems Research* 29, no. 2: 381–400.

Feddersen, M., and L. E. Santana. 2022. "Unpacking the Democratic Affordances of CrowdLaw Concept and Practice." *Politics* 42, no. 3: 340–357.

Fernández Galeote, D., and J. Hamari. 2021. "Game-Based Climate Change Engagement: Analyzing the Potential of Entertainment and Serious Games." *Proceedings of the ACM on Human-Computer Interaction* 5, no. CHI PLAY: 1–21.

Fisher, J. 2020. "Digital Games, Developing Democracies, and Civic Engagement: A Study of Games in Kenya and Nigeria." *Media, Culture and Society* 42, no. 7–8: 1309–1325.

García-Peñalvo, F. J., L. García-Holgado, A. García-Holgado, et al. 2021. "Gaming for Social Inclusion and Civic Participation: The INGAME Project." In *2021 International Symposium on Computers in Education (SIIE)*, 1–4. IEEE.

Gerber, A., M. Ulrich, F. X. Wäger, M. Roca-Puigròs, J. S. V. Gonçalves, and P. Wäger. 2021. "Games on Climate Change: Identifying Development Potentials Through Advanced Classification and Game Characteristics Mapping." *Sustainability* 13, no. 4: 1997.

Gordon, E., and J. Baldwin-Philippi. 2014. "Playful Civic Learning: Enabling Lateral Trust and Reflection in Game-Based Public Participation." *International Journal of Communication* 8: 28.

Hafez, K., A. Skinner, and A. Grüne. 2022. *Foundations of Global Communication: A Conceptual Handbook*. Routledge.

Hahn, B., R. Nithyanand, P. Gill, and R. Johnson. 2016. "Games Without Frontiers: Investigating Video Games as a Covert Channel." In *2016 IEEE European Symposium on Security and Privacy (EuroS&P)*, 63–77. IEEE.

Harris, R., P. Hollins, A. Garg, et al. 2026. "Dialogue Through Play: Games as a Bridge for Citizen Engagement." *Journal of Computer Assisted Learning* 42, no. 3: e70236.

Hesmondhalgh, D., R. Campos Valverde, D. B. V. Kaye, and Z. Li. 2023. "Digital Platforms and Infrastructure in the Realm of Culture." *Media and Communication* 11, no. 2: 296–306.

Isin, E. F. 2008. "Theorizing Acts of Citizenship." In *Acts of Citizenship*, 15–43. Zed Books.

Jäckle, A., J. Burton, M. P. Couper, and C. Lessof. 2019. "Participation in a Mobile App Survey to Collect Expenditure Data as Part of a Large-Scale Probability Household Panel." *Survey Research Methods* 13, no. 1: 23–44.

Jacobs, R. S., S. Werning, J. Jansz, and J. Kneer. 2020. "Procedural Arguments of Persuasive Games." *Journal of Media Psychology* 33, no. 2: 49–59.

de Jong, J. A., and Z. N. Mneimneh. 2021. "Survey Design and Implementation Considerations in International and Cross-National Research." In *Handbook of International and Cross-Cultural Leadership Research Processes*, 246–267. Routledge.

Kahne, J., E. Middaugh, and C. Evans. 2009. *The Civic Potential of Video Games*. MIT Press.

Keusch, F., B. Struminskaya, C. Antoun, et al. 2019. "Willingness to Participate in Passive Mobile Data Collection." *Public Opinion Quarterly, Opinion Quarterly* 83, no. S1: 210–235.

Korn, M., and A. Voida. 2015. "Creating Friction: Infrastructuring Civic Engagement in Everyday Life." In *Proceedings of the Fifth Decennial Aarhus Conference on Critical Alternatives*, 145–156. Aarhus University Press.

Larkin, B. 2013. "The Politics and Poetics of Infrastructure." *Annual Review of Anthropology* 42: 327–343.

Lee, P. 2020. "Communication Rights in a Divided World." *Ecumenical Review* 72, no. 2: 197–208.

Lin, W. Y., and X. Zhang. 2020. "The Role of Digital Media in China: Participation in an Unlikely Place." In *A Research Agenda for Digital Politics*, 72–82. Edward Elgar Publishing.

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Mattelart, T., Y. Hong, S. Milan, D. K. Thussu, and H. Wasserman. 2024. "International Communication: On the Significance of Borders in the Digital Borderless World." *Communication and the Public* 9, no. 2: 131–142.

Mihailidis, P. 2018. "Civic Media Literacies: Re-Imagining Engagement for Civic Intentionality." *Learning, Media and Technology* 43, no. 2: 152–164.

Ming-Tak Chew, M., and Y. Wang. 2021. "How Propagames Work as a Part of Digital Authoritarianism." *Media, Culture and Society* 43, no. 8: 1431–1448.

Murphy, D. 2024. "From Indy to Ubiquity: Minecraft as Platform and Infrastructure." *New Media & Society* 26, no. 11: 6205–6219.

Neset, T. S., S. Juhola, L. Wiréhn, et al. 2020. "Supporting Dialogue and Analysis on Trade-Offs in Climate Adaptation Research With the Maladaptation Game." *Simulation & Gaming* 51, no. 3: 378–399.

Papacharissi, Z. 2022. "Affective Publics." In *The Oxford Handbook of Digital Media Sociology*, 61–72. Oxford University Press.

Parry, E., E. Farndale, C. Brewster, and M. J. Morley. 2021. "Balancing Rigour and Relevance: The Case for Methodological Pragmatism in Conducting Large-Scale, Multi-Country and Comparative Management Studies." *British Journal of Management* 32, no. 2: 273–282.

Plantin, J. C., and A. Punathambekar. 2019. "Digital Media Infrastructures: Pipes, Platforms, and Politics." *Media, Culture and Society* 41, no. 2: 163–174.

Plantin, J. C., C. Lagoze, P. N. Edwards, and C. Sandvig. 2018. "Infrastructure Studies Meet Platform Studies in the Age of Google and Facebook." *New Media & Society* 20, no. 1: 293–310.

Polletta, F. 2018. "Participatory Enthusiasms: A Recent History of Citizen Engagement Initiatives." In *The Participatory Democracy Turn*, 7–22. Routledge.

Richardson, I. 2020. "Ambient Play and Background Gaming: Reflecting on Quotidian Creative Practices." In *The Routledge Companion to Mobile Media Art*, 226–235. Routledge.

Romsdahl, R. J. 2020. "Deliberative Framing: Opening Up Discussions for Local-Level Public Engagement on Climate Change." *Climatic Change* 162, no. 2: 145–163.

Ryan, M., P. Formosa, S. Howarth, and D. Staines. 2020. "Measuring Morality in Videogames Research." *Ethics and Information Technology* 22: 55–68.

Salem, F. 2016. "Open Governance in Authoritarian States." In *Electronic Participation: 8th IFIP WG 8.5 International Conference*, 94–105. Springer.

Shliakhovchuk, E. 2024. "Video Games as Awareness Raisers, Attitude Changers, and Agents of Social Change." *International Journal of Computer Games Technology* 2024, no. 1: 3274715.

Sinpeng, A. 2020. "Digital Media, Political Authoritarianism, and Internet Controls in Southeast Asia." *Media, Culture and Society* 42, no. 1: 25–39.

Szot, J. 2024. "Video Games in Civic Engagement in Urban Planning." *Sustainability* 16, no. 23: 10411.

Tawil-Souri, H. 2015. "Between Digital Flows and Territorial Borders." In *Fragmented Borders, Interdependence and External Relations*, 107–126. Palgrave Macmillan.

Taylor-Smith, E., and C. Smith. 2016. "Non-Public eParticipation in Social Media Spaces." In *Proceedings of the 7th 2016 International Conference on Social Media & Society*, 1–8. Association for Computing Machinery (ACM).

Thiel, S. K., M. Reisinger, K. Röderer, and M. Baldauf. 2019. "Inclusive Gamified Participation: Who Are We Inviting and Who Becomes Engaged?".

Whalen, K., and G. Kijne. 2019. "Game-Based Approaches to Sustainable Innovation." In *Innovation for Sustainability: Business Transformations Towards a Better World*, 375–392. Palgrave Macmillan.

Yau, J. Y. K., D. Kube, H. Drachsler, et al. 2024. "Co-Designing Pilot Games With Citizens and Policy Stakeholders to Increase Climate Action." In *2024 IEEE Gaming, Entertainment, and Media Conference (GEM)*, 1–6. IEEE.

Yuan, E., M. Feng, and X. Liu. 2019. "The R/Evolution of Civic Engagement." *Information, Communication & Society* 22, no. 2: 267–285.

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